

Lecture 0 Review Notes – Probability refresher

CSE5825 – Bayesian Machine Learning

In probabilistic machine learning, we are often interested in estimating the probability of some event. For example,

- $P(\textit{it will snow tomorrow})$
- $P(\textit{a person will develop a disease})$
- $P(\textit{phone call is spam})$

There are two main paradigms in probability and statistics.

Frequentists interpret probabilities as frequencies. For example, we could take the ratio of heads to tails in a million coin flips to determine the probability of heads. Or, to determine if the next phone call is spam, we might observe thousands of phone calls and calculate the spam call frequency.

Bayesians interpret probabilities as quantifying uncertainty (or degree of belief). For example, we believe a coin is fair, and update our belief after a number of coin flips to determine the probability of heads. This distinction will be made more clear in subsequent lectures. In this class, we will adopt a Bayesian perspective to interpret probabilities.

0.1 Monty Hall problem

You find yourself on a game show and are presented with three doors labeled 1, 2, 3. A single prize is placed hidden behind one of the doors and a single goat behind each of the other two. You select a single door and the host opens one of the other two doors revealing a goat. Assuming you value the prize higher than the goat, should you switch your chosen door to the unopened door? Stick with your original pick? Or conclude that both options are equally probable to obtain the prize? We will revisit this problem later in the lecture.

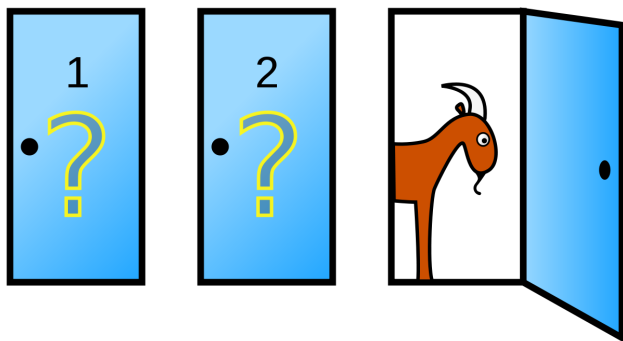


Figure 1: Monty hall problem. Source: Wikipedia.

0.2 Random variables

Random variables describe a random event that has not been observed yet. For example

- the number of tails in 10 flips of a coin
- the time until class ends
- the number of Trump tweets today

Note 1. It is also useful to think of deterministic events as random variables. E.g., the frequency of the word “computer” in published articles.

Random variables express values in a *sample space*. This space can either be *discrete* or *continuous*.

Examples of sample spaces include

- Coin flips: {Heads,Tails}
- Temperature (Kelvin): positive reals, $\mathbb{R}^+$
- Age: positive integers, $\mathbb{Z}^+$

The values that discrete random variables can take are referred to as *atoms*. Random variables are usually denoted with a capital letter. For instance, if X is a random variable denoting a coin flip with sample space $\{H, T\}$, then $P(X = H)$ would be the probability that the coin flip resulted in heads.

Discrete probability distributions assign probability to every atom in the sample space. Keeping with the example of X being a coin flip, then the discrete probability distribution of a fair coin would be

$$P(X = H) = 0.5 \tag{1}$$

$$P(X = T) = 0.5 \tag{2}$$

If we were concerned with characters in documents, we might be interested in the distribution of English characters (Figure 2).

Probabilities over the sample space must sum to one $\sum_x P(X = x) = 1$.

An *event* is a subset of atoms and the probability of an event is the sum of the probabilities of its atoms. For example, on a standard 6 sided die, let D denote the random variables of a single die roll. Then $P(D > 2) = P(D = 3) + P(D = 4) + P(D = 5) + P(D = 6) = 4/6$.

Consider Figure 3 A. The sample space is all of the points (atoms) in the box. The probability function over the sample space is $p(\omega)$ and sums to one over all atoms $\sum_{\omega \in \Omega} p(\omega) = 1$. If the random variable is defined by 10 independent coin tosses, then the atoms are configurations of ten

i	a_i	p_i		
1	a	0.0575	a	■
2	b	0.0128	b	■
3	c	0.0263	c	■
4	d	0.0285	d	■
5	e	0.0913	e	■
6	f	0.0173	f	■
7	g	0.0133	g	■
8	h	0.0313	h	■
9	i	0.0599	i	■
10	j	0.0006	j	■
11	k	0.0084	k	■
12	l	0.0335	l	■
13	m	0.0235	m	■
14	n	0.0596	n	■
15	o	0.0689	o	■
16	p	0.0192	p	■
17	q	0.0008	q	■
18	r	0.0508	r	■
19	s	0.0567	s	■
20	t	0.0706	t	■
21	u	0.0334	u	■
22	v	0.0069	v	■
23	w	0.0119	w	■
24	x	0.0073	x	■
25	y	0.0164	y	■
26	z	0.0007	z	■
27	—	0.1928	—	■

Figure 2: The discrete distribution over the English alphabet for a randomly selected character in a Linux manual. Figure from David JC MacKay. *Information theory, inference and learning algorithms*. Cambridge university press, 2003.

heads or tails. The probability of an atom is the probability of obtaining that particular value or configuration. For instance, the probability of obtaining atom $TTTTTTTTTT$ is 2^{-10} . An event is a the subset of atoms, for example, the number of coins showing heads is 5.

Being more precise, a random variable is a function mapping the sample space (atoms or events) to the reals (a probability). The distribution of a random variable is determined by the probability measure on the sample space.

- Sample space Ω : Set of outcomes. Complete description of the state after experiment.
- Events $\mathbb{F}$: a set whose elements $A \in \mathbb{F}$ are subsets of Ω .
- Probability measure: A function $P : \mathbb{F} \rightarrow \mathbb{R}$ that satisfies
 - $P(A) \geq 0, \forall A \in \mathbb{F}$
 - $P(\Omega) = 1$
 - If $A_1, A_2, \dots$ are disjoint events then $P(\cup_i A_i) = \sum_i P(A_i)$

We are deliberately omitting measure theoretic considerations here.

A *joint distribution* of a collection of random variables is a distribution over the particular configuration of the random variables. In this class, graphical models enable the representation of families

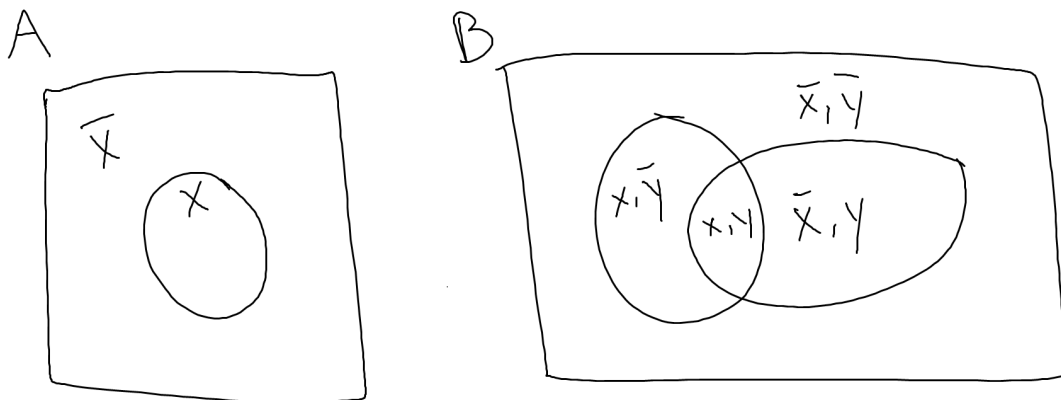


Figure 3: A: Samples space with events x and $\bar{x}$. B: Sample space with two events, x and y .

of probability distributions using graphs and we are often interested in the joint distribution of the random variables and the data.

If we have three coins, then the joint distribution is over the space of all outcomes. In this example, $P(HHH) = P(HHT) = P(HTH) = P(HTT) = P(THH) = P(THT) = P(TTH) = P(TTT) = 1/8$. What are examples of atoms? What is an example of an event?

A *conditional distribution* is the distribution of a random variable when the values of other random variables is known. The conditional distribution of X given Y is denoted $P(X|Y)$ and for particular values, $P(X = x|Y = y)$.

For example,

$$P(\text{we have class Tuesday/Thursday at 3:30pm}) = 0.95 \quad (3)$$

$$P(\text{we have class Tuesday/Thursday at 3:30pm}|\text{snowing}) = 0.75 \quad (4)$$

$$P(\text{we have class Tuesday/Thursday at 3:30pm}|\text{sunny}) = 0.99 \quad (5)$$

The conditional probability of event x given y is

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} \quad (6)$$

Consider Figure 3 B. The conditional probability $P(X = x|Y = y)$, is the relative probability of $X = x$ in the space of $Y = y$.

Chain rule The chain rule can be defined by using properties of conditional probabilities.

$$P(X, Y) = P(X, Y) \frac{P(Y)}{P(Y)} = P(X|Y)P(Y) = P(Y|X)P(X) \quad (7)$$

The general form of the chain rule is

$$P(X_1, \dots, X_N) = \prod_{n=1}^N P(X_n | X_1, \dots, X_{n-1}) \quad (8)$$

Marginalization Graphical models will define a collection of random variables but we are often not interested in a subset. E.g., for random variables A , B , and C , the joint distribution is specified as $P(A, B, C)$. But if we are not interested in B and C , we can compute the *marginal* of X , or, said in a different manner, marginalize out B and C .

$$P(A) = \sum_b \sum_c P(A, B = b, C = c) \quad (9)$$

Consider Figure 3 B. Marginalizing out Y , $P(X = x) = \sum_{y \in Y} P(X = x, Y = y)$, amounts to summing the distribution over the set of configurations where $X = x$.

Bayes rule We can derive *Bayes rule* from the chain rule and marginalization.

$$P(Y|X) = \frac{P(X|Y)P(Y)}{\sum_y P(X|Y = y)P(Y = y)} \quad (10)$$

An example from Scientific American. “A patient goes to see a doctor. The doctor performs a test with 99 percent reliability—that is, 99 percent of people who are sick test positive and 99 percent of the healthy people test negative. The doctor knows that only 1 percent of the people in the country are sick. Now the question is: if the patient tests positive, what are the chances the patient is sick?”

Initially, you might think the probability is 99%. Let’s use Bayes theorem. Let $P(X = x)$ be the probability disease status x and $P(Y = y)$ the probability of a the test being y . We know $P(Y = +|X = +) = 0.99$ and $P(Y = -|X = -) = 0.99$. Only 1% of the country is sick, so $P(X = +) = 0.01$. The probability that the patient is sick given that they tested positive is $P(X = +|Y = +)$.

$$P(X = +|Y = +) = \frac{P(Y = +|X = +)P(X = +)}{P(Y = +)} = \frac{0.99 * 0.01}{(0.99 * 0.01 + 0.01 * 0.99)} = 0.5 \quad (11)$$

Would you use this test in your hospital?

We now revisit the Monty Hall problem. Without loss of generality, assume you select door number 1 initially. Let A_x be the event that the host opens door x and B_x be the event that the prize is located behind door x . Conditional on the host opening door 2, what is the probability the prize

is behind the door if you switch?

$$\begin{aligned}
 P(B_3|A_2) &= \frac{P(A_2|B_3)P(B_3)}{P(A_2)} \\
 &= \frac{P(A_2|B_3)P(B_3)}{P(A_2|B_1)P(B_1) + P(A_2|B_2)P(B_2) + P(A_2|B_3)P(B_3)} \\
 &= \frac{1 * 1/3}{1/2 * 1/3 + 0 + 1 * 1/3} = 2/3
 \end{aligned}$$

For the sake of completeness, the staying strategy yields

$$\begin{aligned}
 P(B_1|A_2) &= \frac{P(A_2|B_1)P(B_1)}{P(A_2)} \\
 &= \frac{P(A_2|B_1)P(B_1)}{P(A_2|B_1)P(B_1) + P(A_2|B_2)P(B_2) + P(A_2|B_3)P(B_3)} \\
 &= \frac{1/2 * 1/3}{1/2 * 1/3 + 0 + 1 * 1/3} = 1/3
 \end{aligned}$$

Independence Random variables X and Y are *independent* if knowing about X has no influence on Y .

$$P(Y|X) = P(Y) \quad (12)$$

One implication of this is

$$X \perp\!\!\!\perp Y \Leftrightarrow P(X, Y) = P(Y|X)P(X) = P(X|Y)P(Y) = P(X)P(Y) \quad (13)$$

Note 2. This is a property of a probability function on a sample space. We will exploit this to factorize the joint probability of graphical models.

Examples of independent variables:

- consecutive rolls of a die
- eye color and favorite type of ice cream

Examples of dependent variables:

- level of education and income
- political affiliation and state of residence

Conditional independence Variables X and Y are *conditionally independent* given Z if

$$P(X|Y, Z = z) = P(X|Z = z) \quad (14)$$

for all values of z . Note that this implies factorization

$$P(X, Y|Z = z) = P(X|Z = z)P(Y|Z = z) \quad (15)$$

for all values of z .

Next section. Next, we will cover graphical models which are a mathematical formalism for reasoning about families of joint distributions. We say *family* here because a single graphical model can yield multiple joint distributions depending on prior probability assumptions. The most important concepts moving forward are **random variables, joint distribution of a collection of random variables, conditional distributions, the chain rule, marginalization, independence, and conditional independence.**

References

MacKay, David JC. *Information theory, inference and learning algorithms*. Cambridge university press, 2003.